

Evidence-Bounded Visual Multimodal Privacy Protection for Egocentric Personal Images

Arun Narayanan
A00048672

arun.narayanan2@mail.dcu.ie

Vaseekaran Krishnan Vinodhan
A00050240

vaseekaran.krishnanvinodhan2@mail.dcu.ie

School of Computing, Dublin City University (AI)
Dublin, Ireland

Supervised by Dr. Liting Zhou, School of Computing, Dublin City University.

Abstract: Egocentric personal images may expose faces, readable text, and screens within the same frame, but privacy methods are often evaluated as separate tasks. This paper presents an evidence-bounded visual multimodal framework that combines detection, protection, evaluation, and objective-aware routing. Experiments use frozen, manually reviewed protocols derived from CASTLE. Combined-risk presence detection reached 0.9804 recall on the held-out multimodal split, although localisation errors and residual text readability remained. Face anonymisation experiments also showed that no evaluated method was uniformly best across objectives and conditions. The Objective-Aware Privacy Router (OAPR) therefore selects only methods that satisfy privacy, utility, visual-safety, runtime, and failure constraints, with deterministic fallbacks where required. The framework supports reproducible privacy-risk reduction for egocentric image sharing, but does not claim complete anonymisation, complete semantic privacy, or production readiness.

Index Terms: Egocentric vision, personal image privacy, visual multimodal privacy, face anonymisation, text and screen redaction, objective-aware routing.

I. INTRODUCTION

A. Privacy Risk in Egocentric Images

Wearable and first-person cameras capture useful records of activities, surroundings, and interactions. They also collect information outside the wearer’s intended subject. One image may contain a bystander’s face, personal text, and content displayed on a phone or computer screen. CASTLE illustrates the scale and diversity of current egocentric collections [1]. EgoPrivacy further shows that sensitive personal and situational information can remain inferable even when a face is unavailable [2]. In this paper, *visual multimodal privacy* means protecting several privacy surfaces within the same image, principally faces, text, and screens. It does not refer to the fusion of different sensor types.

B. Technical and Research Gap

Egocentric capture makes each privacy surface difficult to process. Faces vary in size, pose, blur, occlusion, and illumination. Text can be small, oblique, or only partly readable. Screens introduce reflections, borders, and complex visual content. Missing a sensitive region leaves direct exposure, but unnecessarily strong redaction can reduce scene utility. Existing

de-identification research documents this conflict between privacy and utility [3], while egocentric screen detection remains a specialised task [4]. Most studies therefore optimise one component or one method family in isolation. They do not establish how several protection stages should be evaluated and combined when their accuracy, reliability, visual quality, and compute costs differ.

C. Framework and Research Questions

This work treats privacy protection as an evidence-supported selection problem. CASTLE provides the evaluation setting, while the research questions address egocentric images generally. Reviewed protocols compare detectors, deterministic and generative face protection, joint face, text, and screen processing, and residual failures. OAPR then applies privacy floors, method eligibility, visual-safety checks, runtime evidence, and fallbacks before selecting an action. The study asks:

- RQ1** Which face-detection approaches provide the greatest robustness on egocentric images under practical compute constraints?
- RQ2** How do deterministic and generative face-anonymisation methods compare in privacy protection, image utility, visual quality, runtime, and reliability?
- RQ3** To what extent can an integrated pipeline jointly detect and protect faces, readable text, and screens while preserving the utility of egocentric images?
- RQ4** How does adaptive method selection based on image conditions and privacy objectives compare with using a fixed anonymisation strategy?

D. Contributions

The paper contributes three connected outcomes. First, it provides reviewed and separated protocols for detector, anonymisation, and visual multimodal evaluation. Second, it compares eligible privacy methods using direct metrics and observed failure behaviour, including visual checks that prevent a high numerical score from determining deployment alone. Third, it integrates these findings through OAPR, which records the selected and applied actions and uses a safe fallback when an advanced action is ineligible or fails. Together, these outcomes support an auditable pipeline for risk reduction without treating any single method as universally optimal.

II. RELATED WORK

A. Egocentric Privacy and De-identification

Egocentric cameras record first-person activities and context, but can also expose wearers, bystanders, readable text, screens, and environmental information. CASTLE provides synchronised wearable and fixed-camera views for multimodal understanding, creating challenging imagery for privacy evaluation [1]. EgoPrivacy demonstrates that an egocentric stream can reveal the wearer’s identity, demographics, and situation through cues beyond visible faces [2]. Thapar et al. similarly show that camera motion can encode wearer identity and that suppressing this cue must be balanced against activity-recognition utility [5]. These works establish that privacy in first-person imagery is contextual as well as facial. However, the wider de-identification literature remains dominated by face or person transformation considered separately from text and screens [3].

B. Detection in Difficult Egocentric Conditions

Detection research offers complementary design choices. YOLO formulates object detection as efficient single-network regression [6]; DETR uses global context and direct set prediction [7]; and SCRFD redistributes samples and computation to improve efficient face detection [8]. Their published benchmarks establish general detection or unconstrained face-detection performance, not privacy-weighted reliability on egocentric images. This distinction matters because an undetected face creates privacy exposure, while excessive false positives remove useful scene content. Architecture-level results alone therefore do not identify the best operational policy for this setting.

C. Face Anonymisation

Classical k-Same de-identification groups similar faces and replaces them with an averaged representative, providing an interpretable privacy principle but visibly altering image content [9]. Blur, masking, and related deterministic transformations remain simple and predictable, although stronger settings reduce facial and scene utility. Recent generative work seeks a better balance. FALCO and StyleID manipulate latent representations to suppress identity while retaining facial attributes [10], [11]. RiDDLE adds a diversified, reversible latent encryptor [12]. Diffusion approaches pursue detailed synthesis, attribute preservation, or local control, as illustrated by FAMS, Reverse Personalization, and NullFace [13]–[15].

The methods differ in threat model, reversibility, controllability, required inputs, and evaluation protocol. Their reported results are largely obtained on face-centred benchmarks rather than challenging egocentric imagery. Consequently, strong source-paper scores do not establish reliable transfer to small, blurred, occluded, or non-frontal faces. They also do not capture failed execution or conspicuous visual artefacts. A common target-domain protocol must therefore consider numerical privacy and utility together with runtime, reliability, and direct visual inspection.

D. Text, Screens, and Multimodal Visual Privacy

Scene-text and screen research addresses privacy surfaces that face-only methods leave visible. CRAFT detects character regions and associations, supporting the detection of text with irregular orientations and shapes [16]. Egocentric screen detection has also benefited from multiple views and vision-language reasoning [4]. PrivacyLens demonstrates broader on-device removal of personally identifiable content by combining RGB and thermal sensing [17]. These approaches are valuable but solve different tasks or assume different sensors. They do not by themselves determine how face, text, and screen decisions should be combined within a unified RGB-image privacy policy, nor how residual readable content should be reported after redaction.

E. Privacy, Utility, and Failure Evaluation

Face recognition models provide practical attackers for measuring whether identity remains recoverable. ArcFace supplies discriminative angular-margin embeddings, AdaFace adapts its margin to image quality, and FaceNet provides an established metric-learning baseline [18]–[20]. Such attacker scores measure one privacy channel, not complete protection. Visual utility also requires more than pixel agreement: SSIM measures structural similarity, whereas LPIPS uses deep features to better reflect perceptual judgements [21], [22]. None of these metrics directly captures conspicuous synthesis artefacts, altered gaze or pose, false-positive anonymisation, or residual readable text. This motivates an evaluation that keeps privacy, utility, localisation, runtime, execution failure, and visual safety as distinct evidence before any method is promoted.

F. Research Gap and Positioning

Across the reviewed literature, privacy-weighted face detection, joint face, text, and screen protection, comparable privacy, utility, runtime, and failure evaluation, direct visual review, and gated method selection are not evaluated together under one reviewed egocentric protocol. This work addresses that integration gap by evaluating selected components under shared protocols and connecting their evidence to method eligibility and fallback decisions. The framework remains bounded to the stated protocols and does not claim complete privacy or universal superiority.

III. METHODOLOGY

A. Research Design and Evidence Boundaries

The methodology follows the four research questions through separate reviewed protocols for face detection, face anonymisation, and visual multimodal privacy. Development decisions are separated from held-out evaluation, and every promoted policy is linked to its source metrics, failure evidence, and applied outputs. CASTLE provides the source imagery from which the reviewed experimental protocols are constructed, while conclusions are bounded to the tested protocols and evaluated implementations. The following subsections define the data surfaces, protocol construction, evaluation boundaries and sequence, and method-selection procedure.

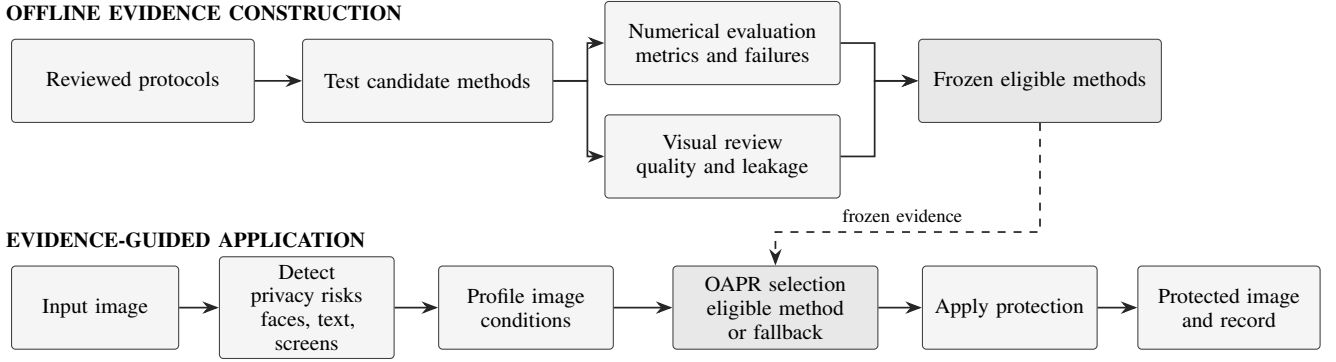


Fig. 1. Two-stage pipeline separating offline method evaluation from evidence-guided protection. Numerical and visual evidence jointly determine method eligibility.

TABLE I
REVIEWED EXPERIMENTAL PROTOCOLS AND EVALUATION BOUNDARIES

Protocol	Images	Ground truth	Use and boundary
Face baseline	500	Reviewed face boxes	Detector development
Egocentric stress	500	Reviewed face boxes	RQ1 stress test; fixed input for RQ2 and RQ4
Visual multimodal	250	116 text and 139 screen boxes	175 development; 75 held-out

Note: The two face protocols contain 2,607 reviewed boxes in total.

B. Source Imagery and Reviewed Protocols

The experimental source comprises 3840×2160 WebP still images from CASTLE, whose time-aligned wearable and fixed-camera recordings span four days and multiple viewpoints [1]. Variation in scale, viewpoint, illumination, motion quality, clutter, text orientation, and screen appearance provides a demanding privacy evaluation setting. The experiments use still images only, not audio, video sequences, or auxiliary sensor streams. Conclusions are limited to the reviewed CASTLE-derived samples rather than cross-dataset generalisation.

Table I separates the protocol roles. Reusing the stress surface as the locked RQ2 and RQ4 input prevents post-result sample changes. The multimodal split keeps configuration and threshold selection separate from final evaluation.

Project reviewers manually verified image-level annotations using dedicated applications. Detector proposals were aids only: reviewers could add, delete, move, or resize boxes before marking an image complete. Corrected boxes form the spatial ground truth, with screen labels taking priority where regions overlap. This joint or sequential review was not a blinded reliability study and does not prove that every semantic or contextual privacy cue was annotated.

C. Threat Model and Evaluation Dimensions

The evaluation treats an unprotected face as a potential source of identity leakage, readable text as disclosure of visible personal information, and a visible display as exposure of screen content. Activities, objects, and surroundings add contextual risk, which the implemented policy does not claim to remove. Protection may also create harm by obscuring non-sensitive scene content or introducing misleading visual artefacts. Accordingly, the evidence separates detection and localisation, face re-identification risk, text readability and screen obscuration, structural and perceptual utility, direct

visual quality, runtime, and execution success. A frame that fails to execute is a recorded failure, not an omitted observation. The outcome is measured privacy-risk reduction under the stated attackers and annotations, not a guarantee of complete anonymisation.

D. Evaluation Sequence and Evidence Controls

Figure 1 summarises the sequence. Fixed manifests identify inputs by path and checksum, while reviewed boxes remain locked across comparable runs. Methods use consistent metrics on their respective protocol surfaces. Per-image privacy, utility, runtime, and success fields ensure that unavailable outputs and failures remain in aggregate evidence. Numerical and visual review are parallel streams because similarity or re-identification scores cannot establish the absence of artefacts, readable content, or false-positive redaction.

Development evidence determines configurations, thresholds, and policies. Only RQ3 has a formal held-out split; other results retain their actual reviewed protocol description. Manifests, annotations, per-image metrics, visual reviews, and selected-output records preserve source-to-table traceability. OAPR uses this frozen evidence after privacy, visual-safety, reliability, runtime, and availability gates. It records the applied action and uses a deterministic fallback when necessary. Scientific materialisation remains distinct from the simplified demonstrator and is not presented as online production deployment.

Evaluation units follow the privacy surface. Face detection is region-level; face anonymisation retains frame-level utility, runtime, and failure evidence alongside face-crop identity evidence. Multimodal evaluation reports both localisation and image-level risk presence. Every attempted image and negative case remains in its denominator, so failures and unnecessary protection affect the results. This prevents successful-only outputs or region-rich frames from distorting conclusions.

E. RQ1 Face-Detection Experimental Design

RQ1 evaluates detector families and policies on the two reviewed 500-image protocols introduced in Table I. A prediction is a true positive only when it is matched one-to-one with an unmatched reviewed face box at intersection-over-union (IoU) ≥ 0.50 . Unmatched predictions and annotations are false positives (FP) and false negatives (FN), respectively. Precision (P), recall (R), and F1 are therefore reported with TP, FP, and FN rather than as isolated scores. Images without an annotated face remain in the protocol: their contribution is assessed through false-positive behaviour and specificity because box recall is undefined. Failed executions also remain in the attempted-image denominator.

Detector configurations are compared using the project-designed privacy-weighted score

$$S_{\text{det}} = 0.65R + 0.25F_1 + 0.10P. \quad (1)$$

Recall has the largest weight because an undetected face can remain directly identifying. F1 retains a balance between missed and unnecessary boxes, while precision penalises avoidable redaction of scene content. Equation (1) supports comparison rather than replacing atomic metrics or defining universal privacy. Deployment also required the project-defined recall floor of $R \geq 0.85$.

F. Detector Comparison and Policy Selection

The evaluation covered RetinaFace, YOLO11Face, and SCRFD stand-alone detectors, agreement fusion, RF-DETR evidence, and learned reranking [6]–[8], [23]. Table II presents representative canonical results on the combined 1,000-image surface.

TABLE II
RQ1 DETECTOR COMPARISON ON 1,000 REVIEWED IMAGES

Method	P	R	F1	S_{det}	Role
RetinaFace-MN-1920	.7252	.8320	.7749	.8070	baseline
Y11-640	.9080	.8972	.9026	.8996	stand-alone
SCR-10G	.9581	.8761	.9152	.8941	precision-led
Y11/SCR agreement	.8877	.9129	.9002	.9072	fusion evidence
RF-DETR/Y11/SCR	.8313	.9321	.8788	.9087	recall-led
RF-DETR reranker	.9075	.9145	.9110	.9129	recall option
Hardened clean LR	.9081	.9175	.9128	.9154	recall variant
Hardened RF	.9319	.9129	.9223	.9172	primary

Notes: Four-decimal metrics. Y11: YOLO11Face; SCR: SCRFD; RF: random forest; LR: logistic regression.

The adopted hardened random forest’s exact verified operating point is:

$$\begin{aligned} P &= 0.9319 & R &= 0.9129 & F1 &= 0.9223 \\ S_{\text{det}} &= 0.917165 & FP &= 174 & FN &= 227 \end{aligned}$$

The RF-DETR Medium evidence run completed all 1,000 images in 119.978 seconds with 850.6 MB peak GPU memory. More aggressive sliced inference increased runtime and false positives and was not promoted.

The score of 0.917165 belongs to the full offline, fold-safe integration of seven detector evidence sources. It is not assigned to the smaller three-source demonstrator, preventing attribution across different implementations.

G. Scene-Condition Profiling

The profiler converts detector output into evidence for downstream policy selection. It combines face count and box geometry with crop scale and quality, image illumination and sharpness, detector confidence, source agreement, and context cues. Its 14 reported labels cover no faces, one face, and multiple faces; small, medium, large, mixed, and very small or distant faces; edge or partial faces; profile or occluded faces; downward egocentric viewpoint; motion blur or low sharpness; dim illumination; and high clutter. Twelve labels have sufficient support for route evaluation. The two rarer scale labels remain reported but are not used as route-eligible evidence.

At execution, the profiler receives detector-produced boxes; reviewed ground-truth boxes are never substituted as runtime inputs. Fixed rules determine face-count labels, a fold-safe multiclass layer assigns mutually exclusive scale labels, and detector telemetry contributes to the contextual labels. This separation prevents ground truth from leaking into runtime features. Route eligibility requires at least 30 reviewed positive instances. Consequently, large-face and very-small-or-distant-face remain descriptive outputs rather than routing evidence.

Condition models are selected using

$$S_{\text{cond}} = 0.50F_{2,\text{macro}} + 0.25F_{1,\text{macro}} + 0.15F_{2,\text{micro}} + 0.10J_{\text{sample}}. \quad (2)$$

Macro F2 gives uncommon, privacy-critical conditions equal influence and favours their recall. Macro F1 checks per-condition balance, micro F2 captures overall recall-sensitive performance, and sample Jaccard measures per-image label-set overlap. This project-designed selection score prioritises recall across uncommon conditions; it is not a universal formula.

On the reviewed 500-image stress protocol, the final hybrid combines fixed count rules, detector telemetry for contextual conditions, and a fold-safe multiclass scale layer. Its aggregate results across the 12 supported labels are:

$$\begin{aligned} S_{\text{cond}} &= 0.925478 & \text{Macro F2} &= 0.933579 \\ \text{Macro F1} &= 0.926823 & \text{Micro F2} &= 0.932814 \\ \text{Sample Jaccard} &= 0.870605 \end{aligned}$$

Performance is condition-dependent. Supported count and common scale labels exceed 0.96 F1. The weaker supported categories are:

$$\begin{aligned} \text{Motion blur or low sharpness:} & & F1 &= 0.8360 \\ \text{Edge or partial face:} & & F1 &= 0.8400 \\ \text{Profile or occluded face:} & & F1 &= 0.8626 \\ \text{High clutter:} & & F1 &= 0.8773 \end{aligned}$$

These weaker categories justify conservative fallback in later routing.

RQ1 finds objective-dependent robustness: hardened random forest provides the strongest balanced point, while alternatives serve precision- or recall-led objectives. Its boxes and condition evidence feed anonymisation and OAPR. CASTLE captures substantial variation in face scale, viewpoint, illumination, blur, occlusion, and clutter. These are recurring challenges in egocentric personal images, making the findings relevant to comparable datasets and capture settings. Exact numerical values remain tied to the reviewed protocols, and OAPR uses conservative fallback when a condition lacks sufficient evidence.

H. RQ2 Anonymisation Protocol and Measures

RQ2 uses the locked 500-image egocentric-stress protocol in Table I, containing 1,284 manually reviewed face boxes. Every method receives the same image manifest and reviewed regions, and all 500 attempts remain in the denominator. An execution without a valid output receives zero privacy, utility, and success credit rather than being removed. Frame-level SSIM and LPIPS measure structural and perceptual preservation [21], [22]; utility U additionally combines face-crop quality, landmark geometry, and background preservation. Identity privacy uses independent AdaFace, ArcFace, and FaceNet attackers [18]–[20]. Lower re-identification rates indicate stronger protection.

For comparable outputs, the project-defined privacy component is

$$P = 1 - \frac{r_{\text{Ada}} + r_{\text{Arc}} + r_{\text{FN}}}{3}, \quad (3)$$

where r_{Ada} and r_{Arc} are per-image rank-1 match rates over valid face crops, and r_{FN} is the proportion exceeding the FaceNet cosine threshold of 0.60. The three-attacker mean is used where all measurements exist; the one missing FaceNet measurement uses the two available attackers, whose rates and denominators remain separately reported. Reviewed no-face frames receive $P = 1$; failures receive $P = 0$. The standardized experimental selection score is

$$S_{\text{anon}} = 0.50P + 0.30U + 0.10R_t + 0.10S, \quad (4)$$

where R_t is constrained runtime suitability and S is execution success. Privacy has the largest weight because recoverable identity is the principal threat. Utility remains substantial because excessive alteration reduces the value of egocentric images; runtime and success penalise impractical or failure-prone methods. These fixed, project-designed weights support method selection only. They are not universal and do not replace the atomic metrics or direct visual review.

I. Deterministic Method Comparison

Table III compares four deterministic transformations. Blur smooths the reviewed face region; pixelation reduces its spatial detail; solid masking replaces it with black; and layered protection combines downscaling, pixelation, blur, and noise. All completed the locked protocol without an execution failure.

TABLE III
DETERMINISTIC ANONYMISATION ON 500 LOCKED IMAGES

Method	Cov.	SSIM	LPIPS	Ada	Arc	FN	t (s)	Role
Blur	500/500	.9929	.0080	.0550	.1158	.0756	.4927	stable
Pixelate	500/500	.9956	.0062	.2971	.4286	.3031	.4858	utility
Solid mask	500/500	.9775	.0195	.0005	.0006	.0270	.4273	privacy-first
Layered	500/500	.9893	.0098	.0136	.0372	.0277	.4410	balanced

Notes: Cov.: successful/attempted images; Ada: AdaFace; Arc: ArcFace; FN: FaceNet; t : mean seconds per image. Re-ID uses attacker-specific decision criteria. AdaFace and ArcFace use 354 face-positive frames; FaceNet uses 353 because one measurement was unavailable. All failure counts are zero; image metrics are means.

The normalised project-score components make the operating trade-offs explicit:

Method	P	U	S_{anon}
Blur	.9419	.7393	.8834
Pixelate	.7581	.7543	.7961
Solid mask	.9921	.5677	.8582
Layered	.9808	.6927	.8898

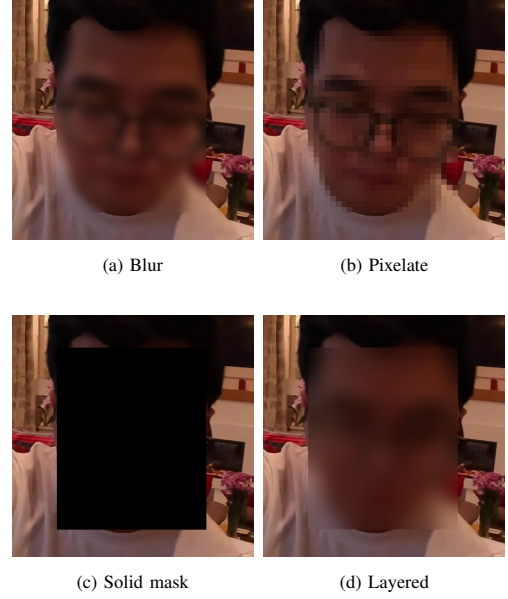


Fig. 2. Deterministic protection outputs for the same reviewed face region from a representative CASTLE frame [1]. The panel is qualitative only; all numerical conclusions use the locked 500-image protocol.

J. Metric and Visual-Review Boundaries

SSIM and LPIPS are computed over the full frame, so their favourable values partly reflect the small image area occupied by a face. They measure preservation, not privacy. Re-identification is restricted to reviewed face-positive frames using the denominators in Table III. Equation (3) is applied per frame before aggregation. These boundaries prevent full-frame similarity or missing attacker evidence from being interpreted as privacy success.

Direct review checks residual identity, region coverage, collateral damage, and conspicuous artefacts. It is not a blinded perceptual study. These non-synthetic methods passed visual-safety eligibility, although protection still depends on correct detection.

K. Interim Answer to RQ2

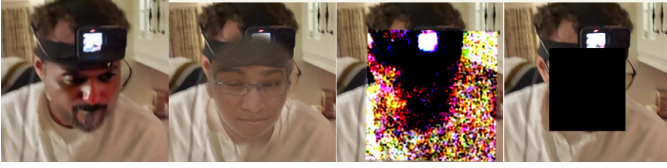
On the locked deterministic comparison, no method is universally best. Layered protection achieved the highest observed composite score on the locked protocol. Solid masking provided the strongest identity suppression, pixelation the highest structural utility, and blur remained the stable practical baseline. Layered combines $P = 0.9808$ with more utility than total masking; solid masking achieves the highest privacy component but the lowest utility. Pixelation leaves substantially more identity evidence. The atomic results explain why the score ordering alone is insufficient: pixelation has the best SSIM and LPIPS but the weakest privacy, whereas total masking produces the best AdaFace and ArcFace suppression at the largest utility cost. Visual review remains decisive because metrics cannot establish the absence of privacy leakage or unacceptable alteration.

The following analysis tests whether advanced methods improve these deterministic reference points without introducing unacceptable failures or visual distortion.

TABLE IV
ADVANCED ANONYMISATION EVIDENCE AND FINAL METHOD ROLES

Method	Cov.	A/R ↓	S/L	t (s)	F	VQ	Role
NF	500/500	.0532/.0837	.9705/.0254	20.389	0	n/q	RO
LD	500/500	.0250/.0360	.9864/.0105	2.296	0	70/100	RO
RP	482/500	.1476/.1768	.9912/.0052	139.440	18	n/q	RO
RD	500/500	.0039/.0016	.9832/.0191	.490	0	14/100	RC
FA	500/500	.0016/.0016	.9826/.0203	16.623	0	22/100	RC

Notes: NF: NullFace; LD: latent diffusion; RP: Reverse Personalization; RD: RiDDLE; FA: FALCO. Cov.: successful/attempted; A/R: AdaFace/ArcFace Re-ID; S/L: SSIM/LPIPS; t : mean seconds/frame; F: failures; VQ: obvious artefacts/reviewed; RO: research only; RC: research comparator. n/q: visually reviewed, not measured in the fixed 100-output artefact audit. RP metrics: successful outputs.



(a) NullFace (b) RiDDLE (c) FAMS (d) Solid mask

Fig. 3. The same reviewed CASTLE face region across three advanced methods and a deterministic fallback [1]. NullFace alters facial geometry, RiDDLE changes pose and gaze, FAMS introduces severe corruption, and solid masking sacrifices utility predictably. The panel is qualitative only.

L. Advanced Method Evaluation

All methods retained the fixed manifest, reviewed regions, common attackers, image metrics, runtime accounting, and failure-inclusive reporting from the RQ2 protocol. NullFace used localised training-free synthesis, while the low-step latent-diffusion branch used region-conditioned inpainting. Reverse Personalization optimised a personalised latent representation [14], [15], [24]. RiDDLE applied invertible identity encryption, FALCO optimised a FaRL-guided latent code, StyleID used a StyleGAN-based identity representation, and FAMS used reference-conditioned diffusion [10]–[13].

Table IV reports the atomic results once. StyleID and FAMS are excluded because neither completed the comparable 500-image metric protocol; their pilot evidence is reported separately. The retained methods have the following operational interpretation:

- *NullFace* completed the locked protocol and produced meaningful identity suppression, but malformed facial geometry and slow execution blocked default promotion.
- *Latent diffusion* preserved full-frame structure, but frequent severe synthesis artefacts restricted it to research evidence.
- *Reverse Personalization* improved after targeted retry, yet persistent failures, appearance drift, weak hard cases, and the slowest runtime prevented operational use.
- *RiDDLE* achieved the highest observed advanced-method composite, but pose and gaze changes and its recorded 80-GB accelerator boundary excluded it from default routing.
- *FALCO* gave the strongest mean two-attacker identity suppression among advanced methods, but visual failures and runtime limited it to a research comparator.

StyleID and FAMS both failed to cross the final pilot visual-quality gate into the locked 500-frame comparison. StyleID retained scale, placement, pose, and compositing failures after its tuned 25-frame pilot. FAMS had broader method-specific development evidence, but its repaired pilot and author-recipe checks retained severe coloured corruption with the obtainable checkpoint.

M. Why Metric-Only Selection Fails

Full-frame SSIM and LPIPS are dominated by unchanged scene area, while low Re-ID can reward any identity change, including an implausible synthetic face. Direct review therefore tested facial geometry, gaze, pose, expression, blending, and collateral alteration. It exposed altered gaze and expression, warped geometry, hard compositing boundaries, unrealistic synthesis, and synthetic insertion when a false-positive detector box was anonymised. Figure 3 illustrates the mismatch directly. This structured project review is independent of the numerical selection score, but is not a blinded perceptual user study.

N. Compute and Runtime Boundary

Compute cost is part of method eligibility, not a deferred implementation detail. Table IV reports measured mean end-to-end time; the full NullFace run required 2 h 49 min 55 s, while Reverse Personalization required 139.440 s per successful frame and still failed on 18 frames. RiDDLE and FALCO were batch-evaluated on an 80-GB CUDA accelerator, peaking at 49.86 and 47.74 GiB respectively, so their timings are device-tied feasibility evidence rather than laptop throughput claims. The later OAPR and App analysis uses CUDA and VRAM tiers, measured same-device ETA, and safe method step-downs.

O. Final Answer to RQ2

No deterministic or advanced method was universally best on the reviewed protocols. Advanced synthesis sometimes improved identity suppression or the numerical privacy-utility trade-off, but visual acceptability, runtime, coverage, and execution reliability changed the operational conclusion. RiDDLE remains the strongest numerical research comparator and FALCO a privacy-strong comparator, not default routes. Layered protection remains the balanced default that passed the visual-safety gate, solid masking the privacy-first fallback, and blur the stable practical baseline within the evaluated framework. The next analysis extends protection beyond faces to text and screens.

IV. RQ3: INTEGRATED VISUAL MULTIMODAL PRIVACY

Protecting faces alone leaves names, messages, documents, and displayed content exposed. RQ3 therefore combines the established RQ1 face boxes with independently reviewed text

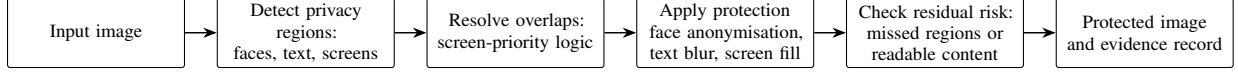


Fig. 4. RQ3 processing sequence. The three privacy surfaces are detected together, overlaps are resolved, protection is applied, and remaining risk is recorded.

TABLE V
HELD-OUT RQ3 COMPARISONS ON 75 REVIEWED IMAGES

A. Image-level privacy-risk presence				B. End-to-end redaction policies			
Detected risk	Precision	Recall	F1	Policy	P_{mm}	U	t (s)
Text present	.2766	.8667	.4194	Adaptive policy	.8680	.6634	.667
Screen present	.7917	1.0000	.8837	Fixed blur/fill	.8680	.6634	.836
Text or screen present	.8333	.9804	.9009				
							$S_{\text{mm-redact}}$
							.8320
							.8319

Note: Each panel compares common measures on the same 75 held-out images. In Panel B, t is mean seconds/image. Region localisation is separate; faces use RQ1 evidence.

A. Reviewed Protocol and Detection Stack

The locked protocol contains 250 manually reviewed images, split into 175 for development and 75 held out for evaluation. Its ground truth comprises 116 text boxes and 139 screen boxes. Configuration and thresholds were selected only on development evidence for the core detectors. The later completion campaigns inspected retained held-out residuals and are therefore reported as post-hoc hardening, not a fresh untouched test estimate. The 250-image surface does not re-score faces; instead, the established RQ1 detector supplies face boxes to the integrated pipeline.

CRAFT with the development-selected 4K recall settings generates text proposals [16]. Screens are detected by the selected union of YOLO11 passes at 640 and 1280 pixels. This multi-resolution evidence addresses the scale and partial-view difficulty also observed in specialised egocentric screen work [4]. Screen regions take priority, so overlapping text proposals are removed rather than redacted twice. If YOLO returns no screen, a dense linked text cluster can complete a screen hypothesis. A strict landscape bottom-edge phone rule is considered only when the screen path remains empty. On the retained held-out split, the documented post-hoc hardening sequence reduced screen-presence misses from $5 \rightarrow 1 \rightarrow 0$; loose high-false-positive rules were rejected.

Presence variants are compared using the project-designed score

$$S_{\text{mm-det}} = 0.65R + 0.25F_1 + 0.10P. \quad (5)$$

Recall receives the greatest weight because a missed privacy-positive image can leave content exposed. F1 and precision retain penalties for imbalance and unnecessary redaction. The score selects methods within this project and does not replace the atomic metrics or define a universal privacy measure.

B. Redaction and Residual Evidence

Six deterministic text and screen operator combinations were evaluated using predicted, not oracle, boxes. Development selection routes predicted no-risk images to copy-through, text-only risk to text blur, and screen-containing risk to text blur

and screen evidence. CASTLE remains the experimental source, while the question concerns comparable egocentric personal images more broadly [1].

plus screen fill. Each attempted image retains privacy, utility, runtime, success, and residual-risk fields. The corresponding project score is

$$S_{\text{mm-redact}} = 0.50P_{\text{mm}} + 0.30U + 0.10R_t + 0.10S, \quad (6)$$

where P_{mm} measures text suppression and screen obscuration, U measures retained image utility, R_t is constrained runtime suitability, and S records execution success. In the implementation, $P_{\text{text}} = 0.70P_{\text{OCR}} + 0.30O_{\text{text}}$, and P_{mm} is the mean of the active ground-truth modality scores: P_{text} for text and O_{screen} for screens. A frame with neither annotated risk scores 1.0. These fixed project weights support selection, not a general anonymisation formula.

C. Held-Out Results and Failure Boundary

Panel A of Table V compares image-level presence directly. Screen recall reached 1.0000; environmental lettering and textures reduced text precision. Their union reached $P = 0.8333$, $R = 0.9804$, and $F1 = 0.9009$. The separate region audit remained weaker: text achieved $P = 0.0679$, $R = 0.7600$, and $F1 = 0.1246$; screens achieved $P = 0.4576$, $R = 0.6429$, and $F1 = 0.5347$ at IoU 0.50, with area coverage 0.8543. Neither presence nor coverage proves complete semantic removal.

Panel B compares end-to-end policies. Privacy and utility were equal after rounding; adaptive routing was slightly faster and scored 0.8320 against 0.8319. The near-identical scores show that adaptive routing remained competitive, but do not establish a meaningful numerical advantage. Residual evidence records two text misses, no screen miss, six readable-text cases, one insufficiently obscured screen, and 37 images with utility below 0.50. These categories are non-exclusive. The low-overlap screen case and contextual cues remain explicit limitations.

D. Answer to RQ3

RQ3 finds that joint face, text, and screen protection provides high-recall, auditable visual multimodal risk reduction, while imperfect localisation and semantic removal preclude a complete-anonymisation claim.

V. RQ4: OBJECTIVE-AWARE PRIVACY ROUTER

A single face-protection method cannot optimise identity suppression, retained scene content, runtime, reliability, and visual quality for every egocentric image. OAPR therefore treats selection as a constrained decision over frozen RQ1–RQ3

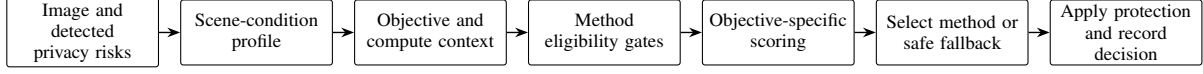


Fig. 5. OAPR runtime decision sequence. Scoring occurs only after evidence and resource gates; an unsupported route terminates in deterministic protection rather than an unverified advanced method.

A. Eligibility Gates

For method m and image evidence x , the project gate is

$$G_m(x) = G_{\text{success}} \wedge G_{\text{privacy}} \wedge G_{\text{quality}} \wedge G_{\text{runtime}}. \quad (7)$$

G_{success} requires available, reliable execution; G_{privacy} enforces the objective’s privacy floor; G_{quality} applies the direct visual findings from RQ2; and G_{runtime} covers measured runtime and compute feasibility. A candidate is eligible only if every required term is true. Thus, a strong Re-ID result cannot promote RiDDLE, FALCO, or another advanced method after it fails the visual or resource boundary. An empty eligible set invokes the supported deterministic fallback. OAPR records rejected methods and reasons, selected and applied actions, runtime, fallback use, and execution failures for later audit.

B. Constrained Selection and Objectives

Selection is restricted to the surviving set:

$$m^*(x) = \arg \max_{m: G_m(x)=1} S_m(x), \quad (8)$$

where $S_m(x)$ combines objective-specific privacy, utility, runtime, and success evidence. Equations 7 and 8 are project-designed rules, not universal formulas.

The materialised research comparison contains seven objective identifiers: privacy first, utility priority, utility under a privacy floor, runtime aware, compute-profile adaptive, failure avoidance, and multimodal risk. Table VI compares their actual 500-image outputs with fixed references on one common metric surface. Four objectives are grouped because they produced identical routed images and metrics, not because their stated purposes are interchangeable.

TABLE VI
FIXED REFERENCES AND MATERIALISED OAPR OUTPUTS ON THE LOCKED 500-IMAGE PROTOCOL

Objective or policy	Applied output	SSIM↑	LPIPS↓	Ada↓	Arc↓	Success	Interpretation
<i>A. Fixed reference policies</i>							
Fixed blur	Fixed blur policy, all frames	.9929	.0080	.0550	.1158	500/500	Balanced baseline
Fixed solid mask	Fixed mask policy, all frames	.9775	.0195	.0005	.0006	500/500	Privacy anchor
Fixed layered	Fixed blur/downscale/noise policy	.9893	.0098	.0136	.0372	500/500	Visual-safe default
<i>B. OAPR objective modes, actual routed outputs</i>							
Privacy first	354 mask, 146 copy	.9793	.0186	.0008	.0008	500/500	Strongest suppression
Failure avoidance	333 layered, 21 blur, 146 copy	.9911	.0090	.0102	.0281	500/500	Reliability-led balance
Runtime aware	275 layered, 79 blur, 146 copy	.9912	.0088	.0164	.0344	500/500	Runtime-sensitive mix
Utility priority / privacy floor / compute adaptive / multimodal risk	354 blur, 146 copy	.9915	.0088	.0258	.0555	500/500	Identical measured output

Note: Panel A repeats the fixed all-frame policies in Table III. Routed rows combine protection with unchanged no-action outputs. Copy denotes an unchanged output where no face-protection action is eligible. Ada and Arc are AdaFace and ArcFace Re-ID rates. All OAPR rows are materialised outputs, not simulations. Across the seven modes, runtime ranged from 0.395 to 0.431 s/image.

C. Interpretation

Privacy-first routing approached the fixed mask’s near-zero Re-ID rate, but retained its visible utility cost. Failure avoidance preserved substantially more image similarity while improving identity suppression over blur. Runtime-aware routing occupied an intermediate operating point. The utility-led group remained close to blur because blur was its strongest eligible fallback. Across all seven modes, mean materialisation runtime ranged from 0.395 to 0.431 s/image and every mode completed 500/500 outputs without a policy failure. These observed trade-offs do not establish global dominance over blur or statistical superiority between neighbouring scores.

The strengthened visual gate fixed the balanced scientific route at 286 layered outputs, 81 solid masks, and 133 reviewed no-face copy-through frames. The four grouped objectives retain different intentions; their identical outputs reflect the eligible methods on this protocol rather than equivalent objective definitions.

D. Visual-Safe Decision and Answer

It completed 500/500 routes with no generative selections, zero failures, and a recorded balanced-objective score of 0.9562. Table VI compares fixed policies with earlier objective-mode routes on the locked face-box surface. The hardened result additionally applies detector-informed safety gates, so its distribution and composite score are reported separately. The gate overrode 13 candidate no-action decisions when high-confidence face evidence remained, reducing 146 initial copy routes to 133 safe copies. These extra protection actions are a utility cost, not execution failures. This explains why direct visual and safety eligibility can supersede an attractive metric-only route. RQ4 therefore finds that OAPR supports auditable, condition-aware and objective-aware selection across the evaluated heterogeneous egocentric conditions. Its contribution is explicit objective selection, exclusion of unsupported or visually unsafe candidates, adaptation to compute and failure evidence, deterministic fallback, and a reviewable decision record, rather than universal metric dominance or complete anonymisation.

VI. RESEARCH DEMONSTRATOR

The implemented App converts the research into an accessible end-to-end privacy workflow. It brings compute-aware planning, face, text, and screen detection, editable review, protection, previews, and audit reporting into one interface. The result is a practical demonstrator of how the verified methods and OAPR evidence can guide real image protection without requiring the operator to understand each model or metric.

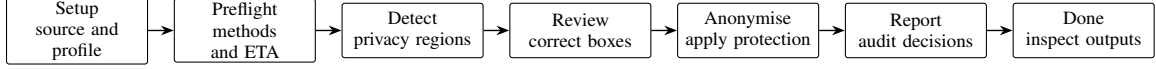
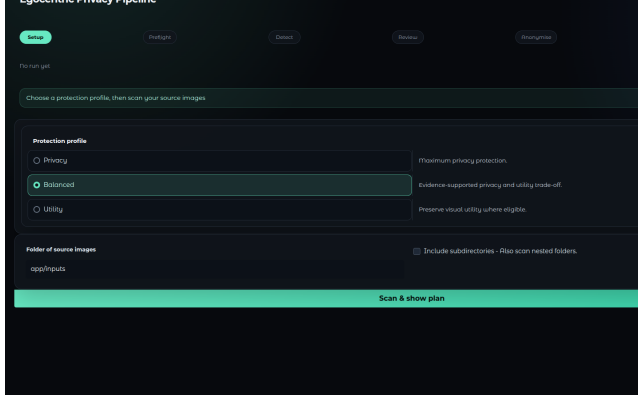
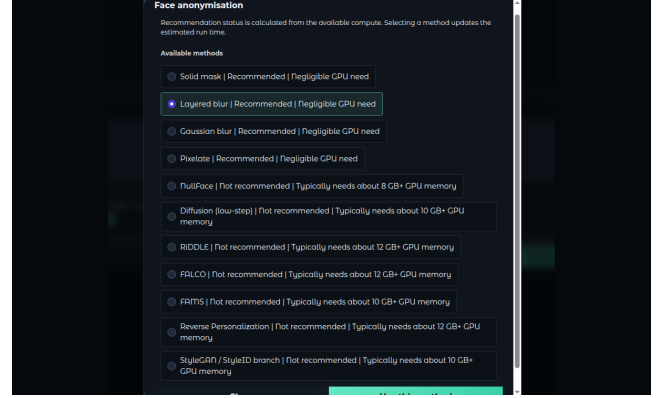


Fig. 6. Operator journey through the App demonstrator. Review is optional, but the remaining stage order is enforced so that method intent and execution records stay aligned.



(a) Setup selects the source and the intended Privacy, Balanced, or Utility profile before any models are loaded.



(b) The anonymisation selector compares deterministic and advanced methods and shows their compute requirements before execution.

A. Guided Preflight

Setup accepts an image folder and a Privacy, Balanced, or Utility profile. Preflight scans the workload, summarises detected risks, presents the available processing stages and method choices, and estimates duration before heavy models load. OAPR evidence informs the proposed methods, while detected CUDA availability and memory determine their compute-fit status. The operator can accept the complete plan or inspect and change any stage.

B. Compute-Aware Protection

After confirmation, the App streams face and visual multimodal detection progress. It selects an evidence-backed detector tier for the available hardware, while low-cost deterministic protection keeps anonymisation practical on lower-compute systems. More capable accelerators can expose the advanced research methods through the same interface. This tiered design allows one workflow to scale from portable hardware to high-memory GPU execution.

C. Reviewer Control and Outputs

The reviewer separates face, screen, and text layers and permits boxes to be added, moved, resized, or deleted before protection. Completion then creates protected images, side-by-side previews, per-image decisions, runtime records, and a readable report. Selected and applied methods, including any safe fallback, remain visible, so a reviewer can follow the complete run from user intent to the final pixel operation.

D. Evidence-Backed Potential

The App operationalises the detector roles, visual-safety decisions, method eligibility, runtime boundaries, and fallbacks established by the OAPR evidence. Recommendations therefore reflect experimental privacy, utility, reliability, visual-quality, and compute findings rather than model popularity alone. The selector makes those choices understandable by showing which methods fit the available hardware before execution begins.

E. Demonstration Value

The demonstrator makes the full framework tangible in one session: it discovers the workload, recommends a feasible plan, exposes alternative methods, supports human correction, protects three privacy surfaces, and produces inspectable outputs and records. Progress indicators and stage navigation make long jobs understandable, while the modular backends allow new detectors and anonymisers to be added without redesigning the operator journey.

F. Limits and Extension

The measured values remain tied to the reviewed protocols, tested methods, and hardware, and residual contextual cues can remain after regional protection. Broader egocentric collections, user studies, and additional devices should be evaluated next. The App's tiered execution, editable review, safe fallbacks, and audit trail provide a strong platform for that extension.

VII. CONCLUSION

This study developed and evaluated an evidence-bounded visual multimodal privacy framework for egocentric personal images. Rather than treating face anonymisation, text redaction, screen protection, visual quality, runtime, and failure handling as separate concerns, the framework connects them through reviewed protocols and an objective-aware selection layer.

RQ1 showed that detector choice depends on the operating objective. The hardened random-forest policy provided the strongest balanced operating point, with $P = 0.9319$, $R = 0.9129$, $F1 = 0.9223$, and a project score of 0.9172, while precision-led and recall-led alternatives retained useful roles.

RQ2 found no universal anonymiser. Advanced methods could improve numerical identity suppression, but execution cost and direct visual artefacts changed their final eligibility. Layered protection remained the balanced visually safe default, with solid masking and blur providing predictable privacy-first and practical fallbacks.

RQ3 extended protection to faces, text, and screens within one RGB-image policy. Combined-risk presence detection reached 0.9804 recall on the held-out split, while region-level and residual-readability audits exposed the remaining localisation and semantic boundary.

RQ4 showed that OAPR can convert these heterogeneous findings into inspectable decisions. Its hardened balanced route completed all 500 outputs with zero execution failures and a recorded score of 0.9562, while safety gates replaced 13 candidate no-action decisions with protection.

The App demonstrates the practical value of this integration. It offers guided preflight planning, compute-aware recommendations, editable multimodal review, fast deterministic operation on modest hardware, advanced options when compute permits, and traceable protected outputs. The principal contribution is thus not a claim that one model solves every privacy condition, but a reproducible way to combine evidence, visual review, compute constraints, and safe fallbacks. Future work should validate the framework on further egocentric collections, additional hardware, video sequences, and broader reviewer studies.

DECLARATION OF GENERATIVE AI USE

Google Gemini was used during the project for coding assistance within Google Colab, including code suggestions and debugging support. Gemini was also used during report preparation to assist with language refinement, LaTeX code, and document formatting. The project README files were generated with Gemini. Its outputs were iteratively reviewed, corrected, and accepted or rejected by the authors.

Gemini did not replace the authors' implementation, manual annotation, experimental execution, visual review, result interpretation, or final judgement. All reported values and claims were checked against the project evidence by the authors. No other generative AI model was used. The authors take responsibility for the submitted work and confirm that they have conformed to the applicable regulations on the use and declaration of Generative AI.

REFERENCES

- [1] L. Rossetto *et al.*, "The CASTLE 2024 dataset: Advancing the art of multimodal understanding," in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 12 629–12 635. DOI: 10.1145/3746027.3758199.
- [2] Y. Li *et al.*, "EgoPrivacy: What your first-person camera says about you?" In *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267, PMLR, 2025, pp. 36 794–36 809. [Online]. Available: <https://proceedings.mlr.press/v267/li25ds.html>.
- [3] W. Khan *et al.*, "Person de-identification: A comprehensive review of methods, datasets, applications, and ethical aspects along with new dimensions," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 7, no. 3, pp. 293–312, 2025. DOI: 10.1109/TBIOM.2024.3485990.
- [4] X. Li *et al.*, "Screen detection from egocentric image streams leveraging multi-view vision language model," *IEEE Transactions on Multimedia*, vol. 28, pp. 4823–4834, 2026. DOI: 10.1109/TMM.2026.3660180.
- [5] D. Thapar *et al.*, "Anonymizing egocentric videos," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 2320–2329. DOI: 10.1109/ICCV48922.2021.00232.
- [6] J. Redmon *et al.*, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 779–788. DOI: 10.1109/CVPR.2016.91.
- [7] N. Carion *et al.*, "End-to-end object detection with transformers," in *Computer Vision – ECCV 2020*, ser. Lecture Notes in Computer Science, vol. 12346, Springer, 2020, pp. 213–229. DOI: 10.1007/978-3-030-58452-8_13.
- [8] J. Guo *et al.*, "Sample and computation redistribution for efficient face detection," in *International Conference on Learning Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=RhB1AdoFfGE>.
- [9] E. M. Newton *et al.*, "Preserving privacy by de-identifying face images," *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 2, pp. 232–243, 2005. DOI: 10.1109/TKDE.2005.32.
- [10] S. Barattin *et al.*, "Attribute-preserving face dataset anonymization via latent code optimization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 8001–8010. DOI: 10.1109/CVPR52729.2023.00773.
- [11] M.-H. Le *et al.*, "StyleID: Identity disentanglement for anonymizing faces," *Proceedings on Privacy Enhancing Technologies*, vol. 2023, no. 1, pp. 264–278, 2023. DOI: 10.56553/popets-2023-0016.
- [12] D. Li *et al.*, "RIDDLE: Reversible and diversified de-identification with latent encryptor," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 8093–8102. [Online]. Available: <https://cvpr2023.thecvf.com/virtual/2023/poster/22988>.
- [13] H.-W. Kung *et al.*, "Face anonymization made simple," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2025, pp. 1040–1050. DOI: 10.1109/WACV61041.2025.00110.
- [14] H.-W. Kung *et al.*, "Reverse personalization," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2026, pp. 988–999. DOI: 10.1109/WACV61042.2026.00103.
- [15] H.-W. Kung *et al.*, "NullFace: Training-free localized face anonymization," in *2026 IEEE 20th International Conference on Automatic Face and Gesture Recognition (FG)*, 2026, pp. 1–10. DOI: 10.1109/FG67764.2026.11557070.
- [16] Y. Baek *et al.*, "Character region awareness for text detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 9365–9374. DOI: 10.1109/CVPR.2019.00959.
- [17] Y. Iravantchi *et al.*, "PrivacyLens: On-device PII removal from RGB images using thermally-enhanced sensing," *Proceedings on Privacy Enhancing Technologies*, vol. 2024, no. 4, pp. 872–891, 2024. DOI: 10.56553/popets-2024-0146.
- [18] J. Deng *et al.*, "ArcFace: Additive angular margin loss for deep face recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 4690–4699. DOI: 10.1109/CVPR.2019.00482.
- [19] M. Kim *et al.*, "AdaFace: Quality adaptive margin for face recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 18 750–18 759. DOI: 10.1109/CVPR52688.2022.01819.
- [20] F. Schroff *et al.*, "FaceNet: A unified embedding for face recognition and clustering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 815–823. DOI: 10.1109/CVPR.2015.7298682.
- [21] Z. Wang *et al.*, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004. DOI: 10.1109/TIP.2003.819861.
- [22] R. Zhang *et al.*, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 586–595. DOI: 10.1109/CVPR.2018.00068.
- [23] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. DOI: 10.1023/A:1010933404324.
- [24] R. Rombach *et al.*, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10 684–10 695. DOI: 10.1109/CVPR52688.2022.01042.